Opgavesæt 1 i vektoranalyse

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1 Exercise 1

Let F be a vector field on $\mathbb{R}^3$ given by:

$$F(x, y, z) = (xy^2z, x^2yz, \frac{1}{2}x^2y^2)$$

calculate the integral:

$$\oint_C \mathbb{F} \cdot dr$$

where $C = \{(x, y, z) \in \mathbb{R}^3 | x^2 + y^2 = 1, z = 4\}$ (for positive orientation in the xy-plane)